

Sample Problems
Lesson: Mathematical Expectation
Second Semester, SY 2025 - 2026

General Instructions: Do as instructed.

1. A server log contains 8 file requests, consisting of 3 video files, 2 image files, and 3 document files. A random sample of 4 file requests is selected. Let X be the number of video files in the sample, and Y be the number of image files in the sample.

(a) Use the marginal densities $g(x)$ and $h(y)$ to determine the following.

- | | | |
|-------------|------------------------|------------------|
| i. μ_x | iii. $\mathbb{E}(X^2)$ | v. σ_X^2 |
| ii. μ_y | iv. $\mathbb{E}(Y^2)$ | vi. σ_Y^2 |

(b) Determine the $\mathbb{E}(XY)$.

(c) Compute the covariance σ_{XY} of X and Y .

(d) Compute the correlation coefficient ρ_{XY} between X and Y .

2. A web server records the response time for user requests in seconds. Let X be the time (in seconds) to process a video file request and Y be the time (in seconds) to process an image file request. Suppose the joint density function of X and Y is

$$f(x, y) = \begin{cases} x + y, & 0 \leq x \leq 1, 0 \leq y \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Use the marginal densities $g(x)$ and $h(y)$ to determine the following.

- | | | |
|-------------|------------------------|------------------|
| i. μ_x | iii. $\mathbb{E}(X^2)$ | v. σ_X^2 |
| ii. μ_y | iv. $\mathbb{E}(Y^2)$ | vi. σ_Y^2 |

(b) Determine the $\mathbb{E}(XY)$.

(c) Compute the covariance σ_{XY} of X and Y .

(d) Compute the correlation coefficient ρ_{XY} between X and Y .

3. A server's response time X (in milliseconds) has a mean of $\mu = 100$ ms and a variance of $\sigma^2 = 25$. Using Chebyshev's Theorem, find

- (a) $P(|X - 100| \geq 10)$
(b) $P(|X - 100| < 10)$
(c) $P(85 < X < 115)$